

Hyperbolic trig functions

Basic skills

$$\text{Q1. a) } \sinh 2 = \frac{e^2 - e^{-2}}{2} \approx 3.63$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\text{b) } \tanh \pi = \frac{e^\pi - e^{-\pi}}{e^\pi + e^{-\pi}} \approx 0.996$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\text{c) } \cosh(-3) = \frac{e^{-3} + e^{-(-3)}}{2}$$

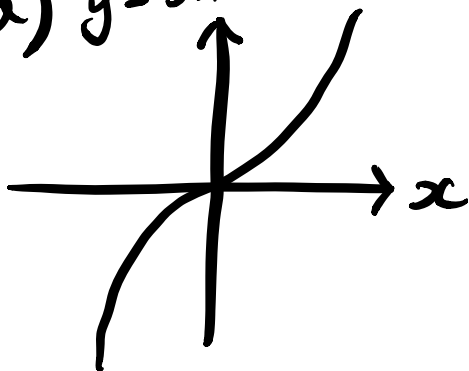
$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$= \frac{e^{-3} + e^3}{2} \approx 10.1$$

same as
 $\cosh 3$

Q2.

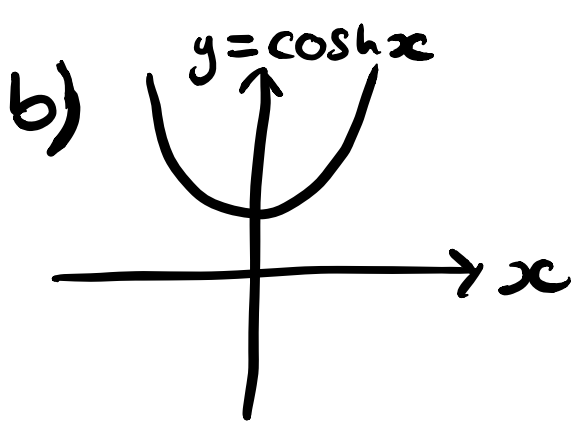
$$\text{a) } y = \sinh x$$



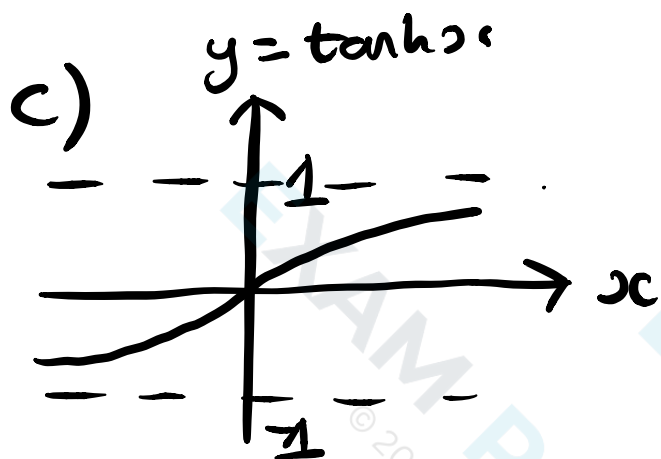
domain: $x \in \mathbb{R}$

all $\uparrow$ real values

range: $y \in \mathbb{R}$



domain: $x \in \mathbb{R}$
 range: $y \in [1, \infty)$
 $y \geq 1$



domain: $x \in \mathbb{R}$
 range: $y \in (-1, 1)$
 $-1 < y < 1$

Q3.

a) Hyperbolic trig graphs do not have waveform graphs so do not have a period so

"Hyperbolic sine has a period of 2π " is False

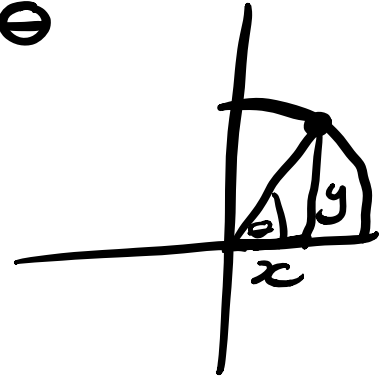
b) Hyperbolic tan doesn't reach 1 or -1 so they aren't max/min just limits so

"The maximum and minimum of hyperbolic tan is $-1/1$ " is False

Q4.

a) As $x = \cos \theta$, $y = \sin \theta$

so $x^2 + y^2 = 1$ unit circle



b) $x^2 - y^2 = 1$

c) $\cosh^2 \theta - \sinh^2 \theta$

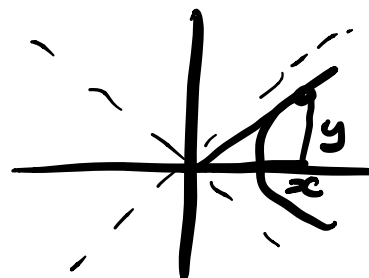
$$= \left(\frac{e^{\theta} + e^{-\theta}}{2} \right)^2 - \left(\frac{e^{\theta} - e^{-\theta}}{2} \right)^2$$

$$= \frac{e^{2\theta} + 2 + e^{-2\theta} - (e^{2\theta} - 2 + e^{-2\theta})}{4}$$

$$= \frac{4}{4} = 1.$$

d) As if $x = \cosh \theta$, $y = \sinh \theta$

then $x^2 - y^2 = 1$



Q5.

a) $\sinh x = 2$

so $\frac{e^x - e^{-x}}{2} = 2 \Rightarrow e^x - e^{-x} = 4$

$\Rightarrow e^{2x} - 4e^x - 1 = 0 \Rightarrow (e^x)^2 - 4(e^x) - 1 = 0$

with $y = e^x$ $y^2 - 4y - 1 =$

$\Rightarrow y = \frac{4 \pm \sqrt{16+4}}{2} = 2 \pm \sqrt{5}$

so $e^x = 2 \pm \sqrt{5} \Rightarrow x = \ln(2 \pm \sqrt{5})$

b) $\cosh x = 6$

so $\frac{e^x + e^{-x}}{2} = 6 \Rightarrow e^x - 12 + e^{-x} = 0$

so $e^{2x} - 12e^x + 1 = 0$

so $e^x = \frac{12 \pm \sqrt{12^2 - 4}}{2} = 6 \pm \sqrt{35}$

so $x = \ln(6 \pm \sqrt{35})$

c) $\tanh x = \frac{3}{4}$

so $\frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{3}{4} \Rightarrow 4e^x - 4e^{-x} = 3e^x + 3e^{-x}$

$\Rightarrow e^x - 7e^{-x} = 0 \Rightarrow e^{2x} - 7 = 0$

$\Rightarrow (e^x - \sqrt{7})(e^x + \sqrt{7}) = 0$

$\Rightarrow e^x = \pm \sqrt{7}$ so $x = \ln(\sqrt{7})$

as $e^x \neq -\sqrt{7}$ as $e^x > 0$ for all x

Q6.

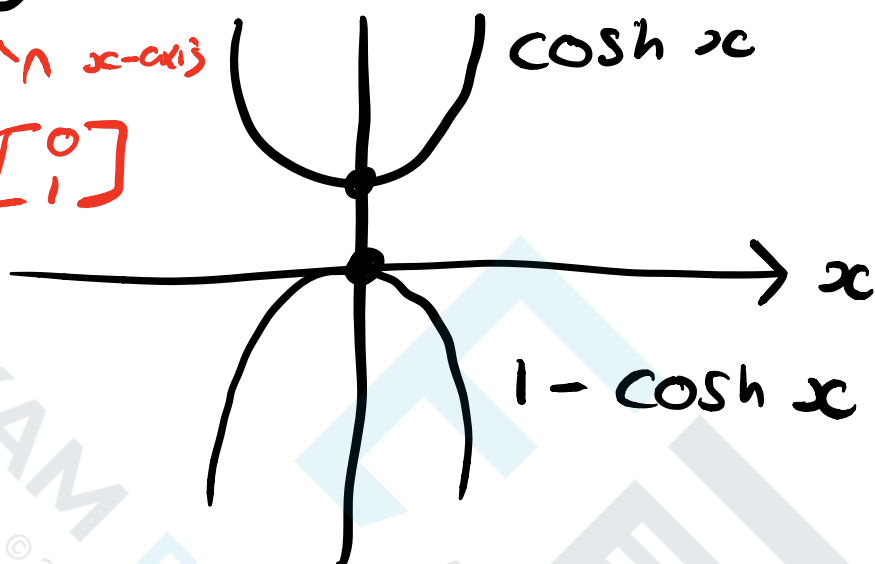
$$\sinh 2x = \frac{e^{2x} - e^{-2x}}{2}$$

$$= \frac{(e^x - e^{-x})(e^x + e^{-x})}{2} = \frac{e^x - e^{-x}}{2} \cdot \frac{e^x + e^{-x}}{2}$$

$$= 2 \sinh x \cosh x = 2 \cosh x \sinh x$$

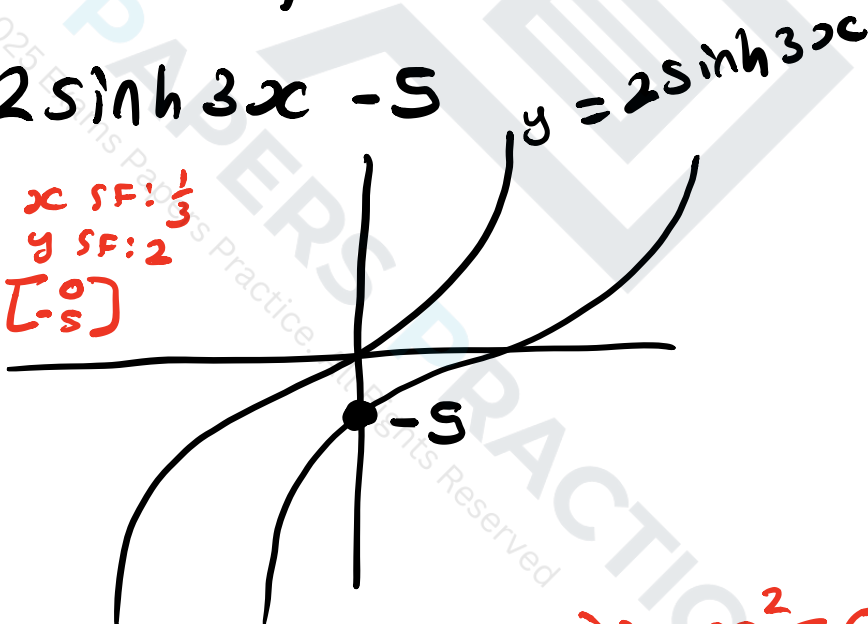
Q7. $y = 1 - \cosh x$

Reflect in x -axis
Translate $[1, 0]$



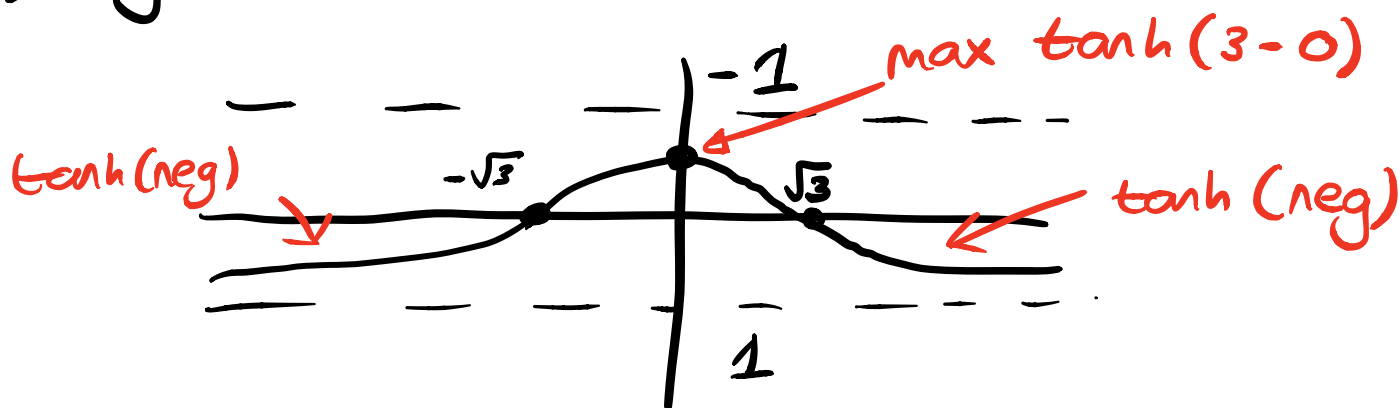
b) $y = 2 \sinh 3x - 5$

Stretch x SF: $\frac{1}{3}$
 y SF: 2
translate $[-5, 0]$



c) $y = \tanh(3 - x^2)$

max is $x^2 = 0$,
as $x \rightarrow \infty$ " $\tanh(-\infty)$ "



Q8. $\lim_{x \rightarrow \infty} \tanh(-x) = \lim_{x \rightarrow -\infty} \tanh(x) = -1$

Q9. $\sinh(0.1) = \frac{e^{0.1} - e^{-0.1}}{2} = 0.1$

$0.1^3 = 0.001$

Bad

$\sinh(1) = \frac{e^1 - e^{-1}}{2} = 1.17$

$1^3 = 1$

okay

$\sinh(5) = \frac{e^5 - e^{-5}}{2} = 74$

$5^3 = 125$

Bad

Not good however graph has similar properties

$\cosh(0.1) = \frac{e^{0.1} + e^{-0.1}}{2} = 1.01$

$0.1^2 + 1 = 1.01$

good

$\cosh(1) = \frac{e^1 + e^{-1}}{2} = 1.54$

$1^2 + 1 = 2$

okay

$$\cosh(s) = \frac{e^s + e^{-s}}{2} = 74.2$$

$$s^2 + 1 = 26 \quad \text{Bad}$$

Again not good although graph looks similar (scale isn't correct)

Q10. $\sinh y = x \quad \frac{e^y - e^{-y}}{2} = x$

$$\Rightarrow e^y - 2x - e^{-y} = 0 \Rightarrow e^{2y} - 2xe^y - 1 = 0$$

$$\Rightarrow e^y = \frac{2x \pm \sqrt{(2x)^2 - 4 \times 1 \times -1}}{2} = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$= x \pm \sqrt{x^2 + 1}$$

but $x - \sqrt{x^2 + 1} < 0$ so only $e^y = x + \sqrt{x^2 + 1}$

$$\text{so } y = \ln(x + \sqrt{x^2 + 1})$$

$$\text{so } \operatorname{arcsinh} x = \ln(x + \sqrt{x^2 + 1})$$